

Education

Rensselaer Polytechnic Institute (RPI), Troy, NY

August 2021 – May 2026

BS in Computational Biology | Lally School of Management MBA, AI and Advanced Computing — GPA: 3.7

Skills

Programming: Python (Pandas, NumPy, Scikit-learn), JavaScript/Typescript, R, SQL/T-SQL

Databases: PostgreSQL, MySQL, Azure SQL, MongoDB, BigQuery, SQL Server (SSMS)

Frameworks & APIs: FastAPI, React/Vite, RESTful services, Node.js (exposure), Flask/Django (exposure)

Cloud & DevOps: AWS (EC2, S3, Lambda, IAM, VPC), Azure, GCP, Git, GitHub, Docker, CI/CD, Linux/Ubuntu, SSH

Dev Tools: VS Code, pgAdmin, Power BI, Tableau, RStudio, Postman, Jupyter, Swagger

Professional Experience

Quanterion Solutions, Utica, NY (Remote)

January 2026 – May 2026

AI Research Consultant

- Built 8-scenario ML buyer intelligence pipeline (Gradient Boosting 99.7% CV, Stacking Ensemble, RF, SVM, KNN; Monte Carlo, permutation importance, K-Means, PCA) classifying defense software modernization urgency; deployed interactive dashboard to GitHub Pages via CI/CD.
- Developed five-step technical debt calculator in HTML/CSS/JS with automated 1–5 urgency scoring and stakeholder-specific output framing; deployed via multi-branch GitHub Actions CI/CD pipeline.

Alithya, Indianapolis, IN (Remote)

October 2024 – August 2025

System Analyst

- Built cloud-native ETL pipelines and queue orchestration systems using Power Automate and Azure SQL, applying Microsoft Copilot for prompt-driven workflow generation and validation, cutting processing errors by 20%.
- Designed scalable Power BI reporting infrastructure to surface claim-error patterns across billing workflows, using Copilot-assisted DAX query generation and pipeline optimization to reduce reconciliation time by 40%.

Motivity Labs/Walmart, Irving, TX

May 2024 – August 2024

Data Analytics Intern

- Developed secure cloud ingestion pipelines connecting Walmart enterprise server infrastructure to Google BigQuery, processing 100GB+ daily call-center datasets to support distributed analytics and operational reporting.
- Optimized Tableau dashboards and warehouse data models with automated refresh pipelines and query performance tuning, cutting manual reporting workloads by 50% and enabling real-time KPI visibility.

Projects

Social Network Graph Database & Operational Data Warehouse, Troy, NY

April 2026 – Present

- Built a SQL Server graph database (210+ nodes, 3 entity types) with MATCH and SHORTEST_PATH traversal queries, and a star schema data warehouse with a cross-database ETL pipeline into four dimension tables and a fact table.
- Implemented the full T-SQL DW lifecycle including staging load, IDENTITY-keyed dimensions, FK-constrained fact table, and SSMS ER diagram, enabling analytical queries on trends, aggregations, and time-based reporting.

NutriTrack – Full-Stack Social Nutrition & Gamification App, Troy, NY

March 2026 – Present

- Designed a 13-table relational schema and production-grade REST API using FastAPI and SQLAlchemy 2.0, with JWT authentication, Pydantic validation, and 51 endpoints across 7 modular routers; deployed via Docker.
- Built background task processing for a gamification engine covering XP awards, streak tracking, level progression, and 17 achievements, backed by an idempotent XP ledger and 92 USDA-sourced seeded foods.
- Currently building the full application in VS Code using AI-augmented development, making all architectural decisions across auth, data modeling, gamification logic, and API design; core backend verified by a 30-assertion smoke test.

Sub-Saharan Cholera Analysis: GDP & Water Sanitation, Troy, NY

September 2023 – March 2024

- Wrangled 15+ years of multi-format public health data (netCDF precipitation, World Bank GDP, shapefile geospatial) using Python (pandas, scipy), resolving format incompatibilities via custom HDF5-to-CSV conversion scripts.
- Applied scikit-learn K-Means and Mean Shift clustering alongside Spearman regression (~0.70) to model cholera outbreak drivers; visualized findings geospatially with geopandas / matplotlib
- Co-authored and presented an ACM ICMHI publication in Yokohama, Japan — DOI: [10.1145/3673971.3673996](https://doi.org/10.1145/3673971.3673996)

Certifications & Extracurriculars

Preparing: AWS Certified Cloud Practitioner | AWS Certified Machine Learning

Pi Delta Psi Cultural Fraternity | RPI Volleyball | Society of Asian Scientists and Engineers | RPI Blockchain club